📊 EAD Variance Analyzer with LangChain

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verify MPOR for netting sets

MPOR Verification for Netting Sets

Overview

The Margin Period of Risk (MPOR) verification compares the system-calculated MPOR values against the values calculated using the SA-CCR methodology embedded in the EAD calculation code.

MPOR Calculation Logic

According to the EAD calculation code, MPOR is calculated as:

MPOR = (2 if margin_dispute else 1) × (20 if is_LNS or is_difficult_replace or is_illiquid else (5 if is_client_facing else 10)) + margin_freq - 1

Where:

- **Base MPOR**: 5 days (client-facing), 10 days (non-client-facing), or 20 days (large netting set/difficult to replace/illiquid collateral)
- **Margin dispute multiplier**: 2x if dispute exists, 1x otherwise
- **Margin frequency adjustment**: Added to base calculation minus 1

MPOR Discrepancies Identified

Netting Set MNA007 shows systematic MPOR discrepancies across all 6 trades:

Netting_ID	Trade_ID	System MPOR	Calculated MPOR	Difference
MNA007	Trade_001	40.0	20.0	-20.0

Netting_ID	Trade_ID	System MPOR	Calculated MPOR	Difference
MNA007	Trade_002	40.0	20.0	-20.0
MNA007	Trade_003	40.0	20.0	-20.0
MNA007	Trade_004	40.0	20.0	-20.0
MNA007	Trade_005	40.0	20.0	-20.0
MNA007	Trade_006	40.0	20.0	-20.0

Analysis

Issue Identified: The system MPOR of 40 days is **double** the calculated MPOR of 20 days for all trades in netting set MNA007.

Potential Root Causes:

1. **Margin Dispute Flag Discrepancy**: The calculated MPOR of 20 days suggests:
    - No margin dispute (multiplier = 1x)
    - Large netting set OR difficult-to-replace trades OR illiquid collateral (base = 20 days)
    - Margin frequency adjustment included

    The system MPOR of 40 days suggests the system may be applying a 2x multiplier for margin dispute, while the calculation logic shows no dispute flag is active.

2. **Inconsistent Margin Dispute Status**: If MNA007 should have margin_dispute = True, the calculated MPOR would be:
    - 2 × 20 + margin_freq - 1 = 40 + margin_freq - 1
    - This would only equal 40 if margin_freq = 1 day

3. **System Override:** The system may be using a different MPOR calculation methodology or applying manual overrides not reflected in the standard SA-CCR formula.

Recommendation

Action Required: Investigate netting set MNA007 to determine:

- Whether margin dispute status should be flagged as True in the input data
- If the system is applying a different MPOR calculation rule for this netting set
- Whether the 40-day MPOR represents a regulatory floor or internal policy override

Impact: While MPOR affects the maturity factor calculation for margined transactions ($MF = 1.5 \times \sqrt{(MPOR/250)}$), the discrepancy does not appear to have caused EAD differences in this case, as the EAD for MNA007 shows zero change between old and new datasets. This suggests the MPOR discrepancy existed in both runs and did not change between versions.

what is the EAD impact for Netting Set MNA007 in old dataset due to MPOR issue.

EAD Impact for Netting Set MNA007 Due to MPOR Issue

MPOR Validation Issue Identified

For Netting Set **MNA007**, there is a significant discrepancy between the System MPOR and Calculated MPOR values across all 6 trades:

Trade ID	System MPOR	Calculated MPOR	Difference
Trade_001	40	20	-20
Trade_002	40	20	-20
Trade_003	40	20	-20
Trade_004	40	20	-20
Trade_005	40	20	-20
Trade_006	40	20	-20

All trades in this netting set show the System MPOR is **double** the Calculated MPOR (40 vs 20 business days).

EAD Impact Analysis (Old Dataset)

Current EAD Metrics for MNA007 (Old Dataset):

- **Replacement Cost**: 9,494.0
- **PFE Add-on**: 1,458,606.55
- **Multiplier**: 1.0
- **EAD**: 2,055,340.77
- **System EAD**: 2,901,185.24

Root Cause

The MPOR directly impacts the **Maturity Factor** calculation in the SA-CCR framework:

- For **margined** netting sets: Maturity Factor = 1.5 × √(MPOR/250)

With MPOR of 40 vs 20:

- **System Maturity Factor** (MPOR=40): $1.5 \times \sqrt{(40/250)} = 1.5 \times 0.4 =$ **0.60**
- **Calculated Maturity Factor** (MPOR=20): $1.5 \times \sqrt{(20/250)} = 1.5 \times 0.283 =$ **0.424**

This represents approximately a **29% reduction** in the maturity factor.

Estimated EAD Impact

Since the Maturity Factor flows through to Trade Add-ons, which aggregate to PFE Add-on, the impact would be:

If System MPOR (40) were used instead of Calculated MPOR (20):

- The PFE Add-on would increase by approximately **41%** (ratio of maturity factors: $0.60/0.424 \approx 1.415$)
- Estimated PFE Add-on with MPOR=40: ~2,064,000
- Estimated EAD with MPOR=40: $1.4 \times (9{,}494 + 1.0 \times 2{,}064{,}000) =$ **~2,902,891**

EAD Impact: The MPOR discrepancy results in an **underestimation of approximately 847,550** in EAD for MNA007 in the old dataset when using Calculated MPOR (20) versus System MPOR (40).

This explains why the System EAD (2,901,185.24) is significantly higher than the Calculated EAD (2,055,340.77) - a difference of **845,844**, which aligns closely with our estimated impact.

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